QP Code : 25900910

MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION NOVEMBER 2025

FIRST SEMESTER

(MDC) - MG1MDCPES101-PHYSICALFITNESSANDHEALTHYLIVING

PARTA

MULTIPLECHOICEQUESTIONS

Answer all questions

Each question carries1 mark

1. WhyisFlexibilityanimportantcomponentofPhysicalFitness?

Ans:**b)Itreducestheriskofinjuryandimproves therangeofmotioninjoints**

2. How does regular physicalactivity affect Mental Health?

Ans: **b) It helps reduce stress and improve mood**

3. Give the correct pathway for air entering the respiratorysystem

Ans: **b) Nose → Pharynx → Larynx → Trachea**

4. What is the best example of the principle of Specificity in action?

Ans: **c) A swimmer focusing on different strokes to improve overall performance.**

5. Howcanafitnessenthusiastbestapplytheprincipleofrecoveryintheir training regimen?

Ans:**b)Includerestdaysandlighterworkoutsessionsbreaks**

6. Which health-related fitness component is often assessed by body fat percentage?

Ans: **b) Body composition**

7. The training which emphasizes quick, explosive movements to enhance neuromuscular coordination?

Ans:**a)PlyometricTraining**

8. Nametheprocessbywhichnutrientsmovefromthedigestivetractintothebloodstreamis called

Ans**:a)Absorption**

9. Aerobics primarily improve which of the following

Ans: **a) Cardiovascular fitness**

10. Which ofthe following is a major risk factor associated with physical inactivity?

Ans: **d) Increased risk of obesity**

PARTB

SHORTANSWERTYPEQUESTIONS

Answer any 5 Questions

Each questioncarries2marks

11. HowcantechnologybeusedtotrackimprovementsinPhysicalFitness?

Ans: Technology helps track fitness progress and improve sports performance in many ways. Devices like fitness trackers, smartwatches, and heart rate monitors provide real-time data onheart rate, effort levels, and training zones, helping athletes train safely and efficiently. GPS tracking measures speed, distance, and pace to improve endurance and technique. Body composition analyzers track fat, muscle, and hydration, helping athletes maintain optimalphysical condition.Sleep and recovery trackers identify poor sleep or fatigue, improvingrecovery and reducing injury risk. Fitness apps and smart gym equipment monitor long-term progress and give personalized training plans. Altogether, technology helps athletes train smarter, recover better, and achieve higher performance in sports

12. Whatarethetwomaincategoriesofphysicalfitness?

Ans: Physical fitness is broadly divided into two main categories: health-related fitness and skill-related fitness.

Health-related physical fitness refers to the components that contribute directlyto good health and daily functioning. These include cardiovascular endurance, muscular strength, muscular endurance, flexibility, and body composition. These components improve the body's ability to perform everyday tasks effectively, reduce the risk of lifestyle diseases, and enhance overall quality of life.

Skill-related physical fitness focuses on qualities that enhance performance in sports and physical activities. This category includes agility, balance, coordination, power, reaction time, and speed. These components are essential for athletes who need quick, precise, and efficient movements to excel in their sport.

13. WhatdoestheacronymFITTstandforinfitnesstraining?

Ans: TheacronymFITTstandsfor:

- F–Frequency:Howoftenyouexercise.

- I–Intensity:Howhardyouexercise.
- T–Time:Thedurationofeachexercisesession.
- T–Type:Thekindofexerciseyouperform.

14. Explaintheprincipleofoptimumloadandhowitcan beappliedtoimprovefitness

Ans: The principle of optimum load states that fitness improves onlywhen the body is exposed to the correct amount of exercise stress. The load should be challenging enough to stimulate adaptation,butnotso highthatitleadstofatigue,overtraining,orinjury.Toolittleload results in no improvement, while too much can harm performance.

To apply this principle, individuals must adjust key training factors such as intensity, frequency, duration,andtype of exercise. Beginners maystartwith light to moderate workouts and gradually increase the load as their body adapts.Athletes may use heart rate zones,resistance levels, or pace targets to maintain an appropriate training load.

Rest and recoveryalso playan important role, as the bodyneedstime to rebuild stronger after each training session. By progressing the load in small, planned steps—known as progressive overload—fitness improvements can be achieved safely and effectively.

15. ExplaintheimportanceofBalanceinphysicalfitness

Ans: Balance is a vital component of physical fitness because it enables the body to maintain stability and control during movement and while at rest. Good balance improves posture, coordination, and overall body awareness. It helps individuals perform everyday activities—such as walking, bending, or carrying objects—more efficiently and safely.

In sports and physical activities, balance plays a key role in enhancing performance. Athletes in disciplines like gymnastics, yoga, martial arts, dance, football, and athletics rely heavily on balance to execute movements accurately and with control. Improved balance also reduces therisk of falls and injuries by allowing the body to react quickly and maintain proper alignment during dynamic movements.

Balance training strengthens the stabilizing muscles of the core, hips, and legs, which supports proper movement patterns and reduces strain on joints. It also enhances neuromuscular coordination, helping thebody respond better to sudden changes in position or direction.

16. Howdoes theregularexercisesreducetheriskofMuscleInjury?

Ans: Regular exercise strengthens themuscles, tendons,and ligaments,making them more capable of handling physical stress. When muscles are strong and flexible, theyare less likely to strain or tear during activity. Exercise also improves blood circulation, ensuring that muscles receive more oxygen and nutrients, which helps them recover faster and stay healthy.

Regular physical activity enhances joint stability and improves coordination, allowing the bodyto move with better control and efficiency. This reduces sudden, awkward movements that oftenlead to injuries. Warm-up and stretching exercises performed regularly also increase the range of motion, preventing stiffness and reducing the chances of muscle pulls. Overall, by improving muscle strength, flexibility, endurance, and neuromuscular coordination, regular exercise significantly lowers theriskof muscle injuries and helps maintain long-termphysical fitness.

17. Whatarethekeycomponentstoincludeinapersonalactivityrecord

Ans: A personalactivity record is used to systematically track daily or weekly physical activities and monitor progress. The essential components include:

1. **TypeofActivity**
   Specifies the form of exercise performed such as walking, jogging, swimming, yoga, strength training, or sports activities.
2. **DurationandIntensity**
   Records how long the activity lasted and the intensity level(light, moderate, vigorous). This helps assess the training load and energy expenditure.
3. **FrequencyofActivity**
   Indicates how often the activity is performed (daily/weekly), helping in planning and maintaining a regular exercise routine.
4. **PerformanceMeasurements**
   Includes measurable data such as distance covered, number of repetitions, sets, speed, calories burned, or heart rate. This helps track improvement over time.
5. **GoalsandTargets**
   Short-term and long-term fitness goals (e.g., improving stamina, reducing weight, building strength) to evaluate progress and maintain motivation.

PARTC

SHORTESSAYTYPEQUESTIONS

Answer any 3 Questions

Each questioncarries5marks

18. Explaintheconceptofphysicalexerciseandhowitdiffersfromphysicalactivity

Ans: Physical activity refers to any movement ofthe bodyproduced bythe muscles that results in energy expenditure. It includes everydaytasks such as walking, climbing stairs, household work, gardening, or playing casually. Physical activity is not always planned or structured; it simply involves moving the body.

Physical exercise, on the other hand, is a planned, structured, and repetitive form of physical activity performed with the purpose of improving or maintaining physical fitness. Exercises are done with specific goals such as increasing strength, endurance, flexibility, or cardiovascular health. Examples include jogging, weight training, aerobics, yoga, and swimming.

Thekey differenceis that while all exercise isphysical activity,notallphysical activity qualifies as exercise. Physical activity is general movement, whereas exercise is purposeful and goal-oriented. Exercise follows principles like frequency, intensity, time, and type (FITT), making it more systematic.

Insummary, physical activity includes all bodymovements, while physical exercise is a deliberate and organized effort to improve fitness and health.

19. Brieflyexplainanythreeprinciplesoffitness

Ans:1)PrincipleofOverload:

The principle of overload states that the body must be exposed to a level of stress greater than what it normally experiences in order to improve fitness. This additional stress forces the bodyto adapt by becoming stronger, faster, or more efficient. Overload can be applied through three main variables:

- Intensity –howhardyouwork(e.g.,liftingheavierweights,runningfaster)
- Duration–howlongyouwork(e.g.,increasingexercisetimefrom20to30minutes)
- Frequency –howoftenyouworkout(e.g.,exercising4daysinsteadof3)

When the body is consistentlychallenged, it responds with positive physiological adaptations such as improved muscle strength, increased aerobic capacity, stronger bones, and better endurance. However, overload must be carefully planned. Excessive overload can lead to overtraining,injury, or burnout, while insufficient overload leads to no improvement. Thus, applying overload in a controlled manner is essential for achieving fitness gains.

2) PrincipleofSpecificity:

The principle of specificity states that training adaptations are specific to the type of exercise performed. This means that the body improves in exactly what it practices. Different activities enhance different components of fitness:

- Endurancetraining(e.g.,running,cycling)improvescardiovascularandrespiratoryefficiency.
- Strength training(e.g.,liftingweights)increasesmusclesizeandpower.
- Flexibility exercises(e.g.,yoga,stretching)enhancejointmobility.
- Skill-basedtraining(e.g.,practicingbasketballdribbling)improvescoordinationandtechnique.

Specificity is crucial for athletes, as training must match the demands of their sport. A sprinter requires explosive power and speed, so their training focuses on short, intense efforts.Amarathon runner focuses on long-duration endurance training. Even within strength training, exercises target specific muscle groups—training the legs improves lower-body strength but does not necessarily improve upper-body fitness. Therefore, a fitness program must be designed according to the individual's goals, sport, or activity.

3) PrincipleofProgression:

The principle of progression emphasizes that the overload placed on the body mustbeincreased gradually over time. Once the body adapts to a particular workout, it is no longer challenging enough to stimulate improvement. To continue progressing, the training load must be raised in small, planned increments.

Progression ensures continuous improvement while reducing the risk of injury that could occur from rapid or sudden increases in intensity. Slow and steady progression allows muscles, joints, bones, and the cardiovascular system to adapt safely.

It is closely connected to the principle of overload: overload creates the stimulus for improvement, and progression ensures that the overload increases at an appropriate pace. Without progression, the body reaches a plateau where no further improvements occur.

20. <u>DescribethePrinciplesofIndividualdifferenceandoptimumload</u>

Ans: The Principle of Individual Difference states that every person responds differently toexercise due to factors such as age,gender, genetics,fitness level,health status, and lifestyle. Thismeans that a training program must betailored to an individual's abilities and needs. What

works for one person may not work for another. Understanding individual differences helps in designing safe, effective, and personalized fitness plans.

The Principle of Optimum Load explains that fitness improves only when the body is given the correctamount of exercise stress—neither too littlenortoo much. The loadshould be challenging enough to stimulate the body to adapt, but not so intense that it leads to fatigue, injury, or overtraining. Applying this principle involves adjusting intensity, duration, and frequency of exercise gradually, ensuring safe and consistent progress.

Together, these two principles help create balanced and personalized training programs that maximize fitness gains while minimizing risks.

21. <u>What are themaincomponentsof Bloodand whatare their functions?</u> Ans:

Blood is made up of four main components, eachwitha specific and vital function:

- RedBloodCells(RBCs):

  Also called erythrocytes, they contain haemoglobin, which carries oxygen from the lungs to the body and returns carbon dioxide back to the lungs for removal. Their main function is gas transport.

- WhiteBloodCells(WBCs):

  Also known as leukocytes, theyare responsible for fighting infections and protecting the body from bacteria, viruses, and other harmful microorganisms. They are essential for immunity.

- Platelets:

  These are small cell fragments that help in blood clotting. When an injuryoccurs, platelets form clots to stop bleeding and help wounds heal.

- Plasma:

  The liquid part of blood that carries nutrients, hormones, waste products, proteins, and other substances throughout thebody. It helps maintain blood pressure, volume, and pH balance.

22. <u>Explain the role of exercise in mitigating the effects of the ageing process. How dodifferent types of exercises (Eg: strength training aerobics) benefit older adults?</u>

Ans: Regular exercise plays a crucial role in slowing down the effects ofageing and helping older adults maintain physical, mental, and functional health. As people age, muscle mass, bonedensity, flexibility, and cardiovascular efficiency naturally decline. Exercise helps counteractthese changes by keeping the body active and strong.

Aerobic exercises such as walking, cycling, and swimming improve heart and lung function,reduce blood pressure, enhance circulation, and lower the risk of chronic diseases like diabetes and heart disease.They also improve stamina and help maintainahealthybody weight.

Strengthtraining isespeciallyimportantfor olderadultsbecauseithelpspreventthenaturalloss of muscle mass (sarcopenia). Lifting light weights or using resistance bands increases muscle strength, improves balance, supports joint health, and reduces the risk of falls and fractures.Italso helps maintain bone density, reducing the chance of osteoporosis.

Flexibility and balance exercises, such as yoga or stretching routines, improve mobility, reduce stiffness, and enhance coordination. These exercises lower the risk of falls and help older adults perform daily activities with greater ease.

Overall, regular exercise enhances physical independence, improves mood, boosts cognitive function, and contributes to a better quality of life for older adults.